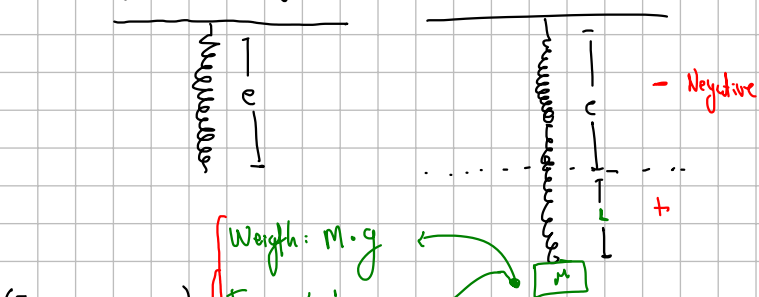


T.D.O 2nd Order

$$\frac{d^2 y}{dt^2} = f(t, y, \frac{dy}{dt})$$

linear with respect to "y" and $\frac{dy}{dt}$

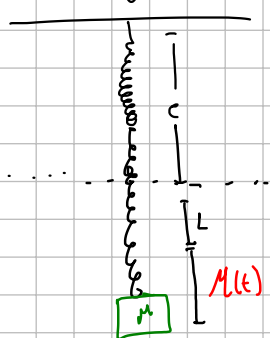
goal = create an equation "M(t)" that give us the position of the spring and mass along the time



Breakdown of your formula: $F_s = -kL$

- F_s : The Restoring Force.
- k : The Spring Constant. This value represents the stiffness of the spring (a higher k means a stiffer spring).
- L : The Displacement (the change in length from the spring's natural state).
- The Negative Sign ($-kL$): This is the most important part for your differential equation. It indicates that the force is always acting in the opposite direction of the displacement. If you pull the mass down, the spring pulls it back up.

Measure the spring position with respect to the time $M(t)$



$\frac{F_s}{m \cdot g = -kL}$

$mg - kL = 0$ (at rest)

Second Newton law = $Ma = F$

Position: $M(t)$

Vel: $\frac{dM}{dt}(t)$

acceleration: $\frac{d^2 M}{dt^2}(t)$

2nd T.D.O = $m \cdot \frac{d^2 M}{dt^2}(t) = F$

Forces Applying to the Mass

1) weight = $M \cdot g$

2) $F_s = -k(L + M(t))$

3) $F_a = -\gamma \frac{dM}{dt}(t)$

4) $F_E = F_E(t)$

$F = Mg + F_s + F_a + F_E$

External stuff (air)

External force = your hand to stretch

$m \cdot \frac{d^2 M(t)}{dt^2} = F(t)$

$m \frac{d^2}{dt^2} (M(t)) = mg - k(L + M(t)) - \gamma \frac{dM}{dt}(t) + F_E(t)$

$m \frac{d^2 M}{dt^2}(t) = -kM(t) - \gamma \frac{dM}{dt}(t) + F_E(t)$

$m \frac{d^2 U}{dt^2}(t) + \gamma \frac{dU}{dt}(t) + kU(t) = F_E(t)$

If we change to m, γ, k depend on t

$m(t) \frac{d^2 U}{dt^2}(t) + \gamma(t) \frac{dU}{dt}(t) + k(t)U(t) = F_E(t)$

Armonic Oscillation
= so fucking hard to solve

But we are gonna solve when they are constants

$m \frac{d^2 M(t)}{dt^2} + \gamma \frac{dM}{dt}(t) + kM(t) = F_E(t)$

Suppose this case

$M=1$

$\gamma=0$

$k=-1$

$F_E(t)=0$

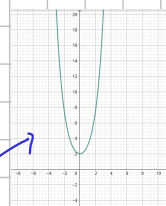
$\frac{d^2 y}{dt^2} - y = 0$

Solutions

$y(t) = e^t$

$y(t) = e^{-t}$

$y(t) = C_1 e^t + C_2 e^{-t}$

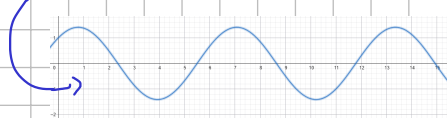


$\frac{d^2 y}{dt^2} + y = 0$

Solutions

$y(t) = \cos(t)$ or $\sin(t)$

$y(t) = C_1 \cos(t) + C_2 \sin(t)$



$m \frac{d^2 M}{dt^2}(t) + \gamma \frac{dM}{dt}(t) + kM(t) = F_E(t)$

We are gonna propose a solution of exp type e^t

$M(t) = e^t$

$\frac{dM}{dt}(t) = e^t$

$\frac{d^2 M}{dt^2}(t) = e^t$

We are gonna replace the corres function $M(t)$... or derivative

$$m \frac{d^2 u}{dt^2}(t) + \gamma \frac{du}{dt}(t) + \kappa u(t) = F_E(t)$$

$$m(r^2 e^{rt}) + \gamma(r e^{rt}) + \kappa(e^{rt}) = 0$$

$$e^{rt}(mr^2 + \gamma r + \kappa) = 0 \quad \left[\begin{array}{l} \text{one of this has to be } 0 \\ e^{rt} \neq 0 \end{array} \right]$$

then

$$mr^2 + \gamma r + \kappa = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$1r^2 + 0r + -1 = 0$$

$$r^2 + r - 1 = 0$$

$$r = \frac{-\gamma}{2m} \pm \frac{\sqrt{\gamma^2 - 4m\kappa}}{2}$$

$$r = \pm 1$$

$$D = \gamma^2 - 4m\kappa$$

D = Discriminant

→ Case 1
→ Case 2
→ Case 3

$$w = \sqrt{D}$$

Case 1

$$r_1 = \frac{-\gamma}{2m} + \frac{w}{2m}$$

$$r_2 = \frac{-\gamma}{2m} - \frac{w}{2m}$$

solutions generals ↓

$$y_1(t) = e^{\frac{1}{2m}(w-\gamma)t}$$

$$y_2(t) = e^{\frac{1}{2m}(w+\gamma)t}$$

This two solutions are gonna generate all the other ones

We are gonna transform the problem into a Linear Algebra

Remember the Key Property: $T(\hat{x} + \hat{y}) = T(\hat{x}) + T(\hat{y})$
(Linear Independency)

$$y_1(t) = e^{\frac{1}{2m}(w-\gamma)t} \quad y_2(t) = e^{\frac{1}{2m}(w+\gamma)t}$$

$$y_g(t) = A e^{\frac{1}{2m}(w-\gamma)t} + B e^{\frac{1}{2m}(w+\gamma)t}$$

Any A or B will give you the solution if there is not initial condition

→ y_g General Solution
We get here by applying Linear algebra

Case 2. $D=0$, $w=0$, $r=-\frac{\gamma}{2m}$

We start knowing that one solution, with this values is $u_2(t) = e^{-\frac{\gamma}{2m}t}$

...
...
Crazy thing...

Case 3. $D < 0$;

$$w = \pm \sqrt{D}$$

$$w = \pm \sqrt{-1|D|}$$

$$w = \pm i\sqrt{|D|}$$

$$x^2 + 1 = 0$$

$$x = \pm \sqrt{-1}$$

$$\hookrightarrow i$$

Complex Number